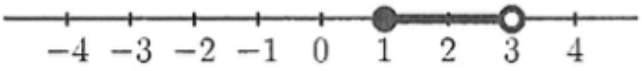
Inequalities

Questions 1–34

Question 1. For each of the following find the values which have been identified, using set notation:

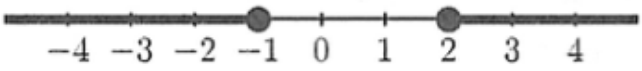
a)



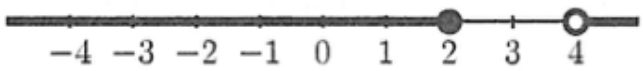
b)



c)



d)



Question 2. For each of the following find the values which have been identified, using interval notation:

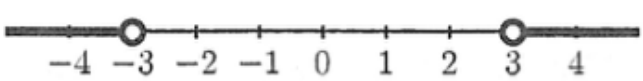
a)



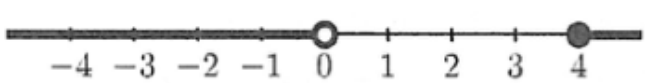
b)



c)



d)



Question 3. Draw a number line to show thes values of real number x which belong in the following sets:

a) $\{x : x \leq -3\} \cup \{x : x > 2\}$
b) $\{x : x < 1\} \cup \{x : x \geq 2\}$
c) $\{x : x \geq -2\} \cap \{x : x < 3\}$
d) $\{x : x > -1\} \cap \{x : x \leq 2\}$
e) $\{x : x > 1\} \cap \{x : x < -3\}$

Question 4. Solve the following inequalities giving your answer in set notation:

a) $3x - 8 < 4$
b) $6 - 2x \leq 15$
c) $6 + 5x > 2 - 3x$
d) $1 + 9x \leq 6 - x$
e) $-2 \leq 3x + 4 \leq 10$
f) $-5 < 5x - 8 < 22$
g) $-1 < 2x + 1 < 1$
h) $-1 < ax + b < 1$ where $a > 0$
i) $-1 < ax + b < 1$ where $a < 0$

Question 5. Solve the following inequalities, writing your answer in interval notation:

a) $\frac{x+14}{3} < 2(x-1)$

b) $\frac{4(x+1)}{3} \geq \frac{x-6}{2}$

c) $\frac{4x+1}{3} - \frac{2x-1}{2} \leq 1$

Question 6. Solve the following inequalities giving your answer in set notation:

a) $(x+2)(x-4) < 0$

b) $(x+1)(x+7) \geq 0$

c) $x^2 + 3x - 10 \leq 0$

d) $x^2 + 5x + 6 > 0$

e) $-(x+3)(x-8) < 0$

f) $(2-x)(4+x) \geq 0$

g) $x^2 + 3x - 1 > 0$

h) $x^2 - 2x + 1 \geq 0$

i) $-x^2 + x - 3 > 0$

Question 7. Solve the following inequalities giving your answer in interval notation:

a) $x^2 - x > 20$

b) $2x^2 < 10 - 8x$

c) $4x^2 + 4x \geq 35$

d) $6x^2 \leq 7x + 5$

Question 8. Freddie tries to solve the inequality $2x - x^2 + 8 \leq 0$. His workings and answer are as follows:

$$x^2 - 2x - 8 \leq 0$$
$$(x + 4)(x - 2) \leq 0$$
$$x = -2, x = 4$$
$$\{x : x < -2\} \cup \{x : x > 4\}$$

a) Identify all his mistakes.

b) Fully correct his workings.

Question 9. Find the set of values of x for which the line $y = 2x + 3$ lies below the curve $y = x^2$ giving your answer in set notation.

Question 10. Find the set of values of x for which the line $y = 19 - 10x$ lies above the curve $y = (2x + 1)^2$ giving your answer in interval notation.

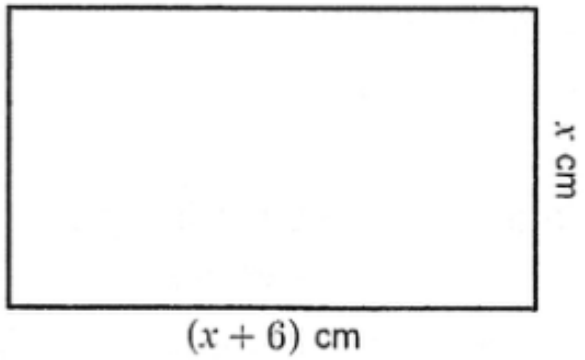
Question 11. Solve $\frac{1}{x^2} < 529$ giving your answer in interval notation.

Question 12. Solve $\frac{1}{x^2} \geq a^2$ where $a \neq 0$. Give your answer in terms of a and in set notation.

Question 13. Re-write the following phrases into mathematical inequalities:

a) x is more than 12
b) x is less than 100
c) x is at most 5
d) x is at least 10
e) x is at least 50% more than y
f) x is at most 20% more than y

Question 14. A picture frame of height x cm is 6 cm wider than it is tall. The width needs to be at least 75% more than its height.



a) Show that $x \leq 8$.

The back of the picture frame has an area of at most 160 cm^2.

b) Find the possible values of x giving your answer in set notation.
c) Hence state its maximum area.

Question 15. Joe wants to install a rectangular swimming pool in his garden. He would like it to be 2 metres longer than it is wide but at least 8 metres long. He wants a pool that is less than 80 m^2 in area. Given the width is x metres, find the possible values of x giving your answer in set notation.

Question 16. Prove that $x^2 + 3x > -4$ for all real values of x.

Question 17. The equation $x^2 + kx + 9 = 0$ where k is a constant has real roots.

a) Show that $k^2 - 36 \geq 0$.

b) Hence find the set of possible values of k giving your answer in set notation.

Question 18. Find the values of k for which each of the following equations have no real roots, giving your answer in set notation:

a) $x^2 + 3kx + 1 = 0$

b) $x^2 - kx + 5 - 2k = 0$

c) $(2k - 1)x^2 + (2k + 3)x + k = 0$, $k \neq \frac{1}{2}$

Question 19. Find the set of values of k such that the curve $y = x^2 - 3kx + 8k$ lies entirely above the x-axis. Write your answer in set notation.

Question 20. Find the set of values of k such that the curve $y = (k+2)x^2 + kx - 2k$ lies entirely above the x-axis. Write your answer in interval notation.

Question 21. Find the set of values of k such that the curve $y = (k+1)x^2 - kx + k$ lies entirely below the x-axis. Write your answer in set notation.

Question 22. Prerequisite knowledge for Q22: Circles Find the values of k such that the line $y = x + k$ intersects the circle with equation $x^2 + y^2 = 16$. Write your answer in set notation.

Question 23. Prerequisite knowledge for Q23: Cubic Curves By first sketching the curve, solve the following inequalities, writing your answer in set notation:

a) $(x-3)(x+2)(x+1) > 0$
b) $(1-x)(x-5)(1+x) > 0$
c) $(x-6)(x-1)(2-x) \leq 0$
d) $x(5-x)(3+x) \geq 0$

Question 24.

a) Sketch the curve $f(x) = x^2(x-2)$ labelling the points where the curve and the axes meet.
b) Hence find the set of values of x such that $f(x) \leq 0$ giving your answer in set notation.

Question 25.

a) Sketch the curve $f(x) = (x+1)(x-3)^2$ labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) > 0$ giving your answer in set notation.

Question 26.

a) Sketch the curve $f(x) = x^2(x-a)$ where $a > 0$, labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) > 0$ giving your answer in set notation.

Question 27.

a) Sketch the curve $f(x) = (x + 2)(x - a)^2$ where $a > 0$, labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) \leq 0$ giving your answer in set notation.

Question 28.

a) Sketching the graph of $y = (a - x)(x - b)^2$ where $a > 0$ and $0 < b < a$, labelling the points where the curve and the axes meet.

b) Find the values of x such that $(a - x)(x - b)^2 \geq 0$ giving your answer in set notation.

c) Find the values of x such that $(a - x)(x - b)^2 \leq 0$ giving your answer in interval notation.

Question 29. Prerequisite knowledge for Q29: Factor Theorem Given $f(x) = 4x^3 - 24x^2 + 21x - 5$,

a) Use the factor theorem to show that $(x - 5)$ is a factor of $f(x)$.
b) Hence fully factorise $f(x)$.
c) Sketch $y = f(x)$.
d) Find the values for which $f(x) \leq 0$ giving your answer in interval notation.

Question 30. Prerequisite knowledge for Q30-Q31: Graph Transformations Given $f(x) = x^3+4x^2-3x-18$,

a) Show that $(x - 2)$ is a factor of $f(x)$.
b) Hence fully factorise $f(x)$.
c) Sketch $y = f(x)$.
d) Solve $f(2x) = 0$.
e) Hence, find the values for which $f(2x) > 0$ giving your answer in set notation.

Question 31. A curve is given by $f(x) = -2x^3 + 18x^2 - 28x - 48$.

a) Fully factorise $f(x)$.

b) Hence, find the values for which $f(\frac{1}{2}x) > 0$ giving your answer in set notation.

Question 32. Prerequisite knowledge for Q32-Q34: Exponentials and Logarithms Solve the following inequalities, writing your answer in set notation:

a) $e^x - 10 \leq 0$

b) $4^x - 5 \geq 0$

c) $(\frac{1}{3})^x > 9$

d) $(\frac{1}{5})^x < 1$

e) $0.4^x > 6.25$

Question 33. Giving your answer in set notation, find the set of values of x for which $2 \times 3^{2x} - 3^x - 28 < 0$.

Question 34. Giving your answer in set notation, find the set of values of x for which $e^x - \frac{4}{e^x} > -3$.